

Crystal-Locked Medium-Wave Transmitter



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Here is a low-power AM transmitter for the medium-wave broadcast band. This circuit is perfect for broadcasting your favourite music around the house. The transmit frequency for the circuit is crystal-locked to 1500kHz.

Circuit and working

Fig. 1 shows the circuit diagram of the crystal-locked medium-wave transmitter. It is built using a 14-stage divider and oscillator CD4060 (IC1), an n-channel MOSFET transistor 2N7000 (T1), low-power audio amplifier LM386 (IC2) and a few other components.

Test Points

Test point	Details
TP0	0V (GND)
TP1	12V
TP2	1500kHz

A 24MHz crystal (X_{24}) is used in conjunction with IC1 to generate the carrier frequency. The crystal frequency is divided

by 16 by IC1 to give the final output frequency of 1500kHz at pin 7 of IC1. The square wave from IC1 drives the class C amplifier built around T1. The MOSFET works as a fast on/off switch. TV balun core RFC2 converts short pulses into a sine wave and also blocks RF from entering audio amplifier IC2. Additional filtering is provided by TV balun core RFC1 and capacitor C4. The amplified signal is coupled to the antenna via capacitor C6.

The AM modulation is achieved by varying the drain voltage of T1. When no audio signal is present, the output of IC2 remains high. The audio modulation changes this voltage during modulation peaks. Potmeter VR1 controls the modulation depth and should be adjusted for maximum modulation without distortion. The audio should be loud and clear on a medium-wave AM receiver.

Use of a different crystal will change the frequency of the transmit-

PARTS LIST

Semiconductors:

- IC1 - CD4060, 14-stage divider and oscillator
- IC2 - LM386 low-power audio amplifier
- T1 - 2N7000 n-channel MOSFET

Resistors (all 1/4-watt, 5% carbon):

- R1 - 1-mega-ohm
- R2-R3 - 1-kilo-ohm
- VR1 - 10-kilo-ohm potmeter

Capacitors:

- C1, C2 - 22pF ceramic disk
- C3-C6 - 100nF ceramic disk
- C7, C8 - 10µF, 25V electrolytic

Miscellaneous:

- CON1 - 2-pin terminal connector
- S1 - On/off switch
- RFC1 - 10 turns, 28SWG, TV balun core
- RFC2 - 50 turns, 28SWG, TV balun core
- X_{24} - 24MHz crystal
- JACK1 - Audio-input jack
- ANT - Wire antenna (around 10m or more)
- Shielded cable

ter. Crystals in the range of 10-24MHz can be used to cover the medium-wave band from 625kHz to 1500kHz.

Construction and testing

An actual-size, single-side PCB layout for the transmitter is shown in Fig. 2 and its component layout in Fig. 3.

Assemble the circuit on the PCB. Keep all leads as short as possible. Use good-quality shielded cable between the audio source and transmitter. Use a 10m or longer piece of wire for the antenna. A 12V hum-free regulated power supply is recommended. ●

Joy Mukherji is an electronics hobbyist and a small-business owner in Albany, New York, USA. His interests include designing radio frequency circuits

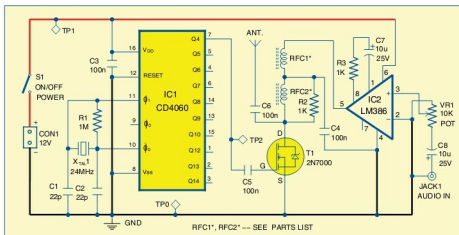


Fig. 1: Circuit diagram of the crystal-locked medium-wave transmitter

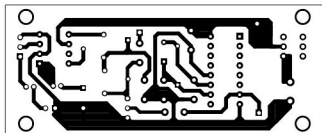


Fig. 2: An actual-size PCB layout of the transmitter circuit

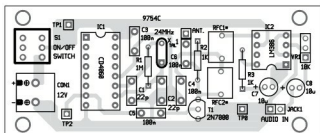


Fig. 3: Component layout of the PCB